Data Architect Transformation Program

Completed

A 12-week, 109-hour pathway that grew the client's data architecture capability from within - helping senior engineers move from building solutions to architecting outcomes.

109 hours | **12** weeks | **11** competency areas | **8+** years' experience cohort | Hands-on, case-study led

As Data and AI initiatives gained momentum, the client set out to grow the architecture leadership that would shape its enterprise-wide data transformation. Partnering with StackRoute | RPS, it chose to invest in its own experienced talent - building a structured route from technical delivery into architecture-led roles.

THE CHALLENGE

- **Scaling architecture leadership** - growing Data and AI demand called for more architects to lead scalable, secure, cloud-native platforms
- **From delivery to design** - experienced developers and tech leads ready to take on architectural decisions and trade-offs
- **Building from within** - a defined internal pathway into architecture-led roles

THE SOLUTION

A single, foundation-first journey. The programme strengthened architecture fundamentals, then layered modern patterns, governance, cloud-native design and DataOps - and brought it all together through real-world case studies and an end-to-end architecture blueprint backed by decision records.

CAPABILITY BUILT

- **Strong architecture foundations** - architecture and modelling principles, SQL/NoSQL and polyglot persistence
- **Pipelines for every data shape** - batch, streaming, CDC and ETL/ELT workflows, orchestrated and automated
- **Trade-offs made explicit** - business vs technical decisions argued in case studies and recorded as ADRs
- **Modern patterns in practice** - Lakehouse, Data Mesh and Data Fabric applied across Azure, AWS and GCP
- **Governance built in, not bolted on** - quality, observability, security, cataloguing and MDM frameworks
- **Designed to run and to last** - fault tolerance, disaster recovery, DataOps CI/CD and FinOps

Each capability maps to a programme objective: scaling architecture leadership, moving experienced engineers from delivery to design, and building that capability from within.

THE NUMBERS THAT MATTER

+2.0
AVG PROFICIENCY GAIN
Across 11 competency areas, developing to competent and proficient

90
NET PROMOTER SCORE
Strong advocacy from an experienced, senior cohort

4.6–4.9
LEARNER SATISFACTION
Consistently high CSAT, rated out of 5 across the journey

109 hrs
OVER 12 WEEKS
Guided independent learning plus instructor-led sessions and labs

THE PROGRAMME IN ONE FRAME

Foundations & modelling
- Data architecture foundations (guided learning)
- Reimagining modern data architecture
- Advanced data modelling and NoSQL vs SQL implementation
- Architecture principles and fault-tolerant design

Patterns & processing
- Modern architecture patterns and design principles
- Data Mesh and Data Fabric integration paradigms
- Serverless and data processing
- Streaming data pipelines with Kafka

Governance & trust
- Data quality, governance and observability
- Security and trust in data projects
- Essentials of MDM in data architecture
- Advanced security and disaster recovery

Cloud, DataOps & value
- Pipeline orchestration (NiFi / Azure Data Factory)
- DataOps and CI/CD for data pipelines
- Cloud-native solutions, monitoring and FinOps
- Visualising and communicating data insights

HOW IT WAS DELIVERED

Format
Blended 12-week journey: guided independent learning, then instructor-led sessions - 109 hours in all.

Approach
Foundation-first - fundamentals, then modern patterns, then real-world trade-offs.

Method
Hands-on labs and case-study sessions on complex datasets, rotating across Azure, AWS and GCP.

Evidence
Knowledge checks, two milestone assessments and a review-and-refactor cycle on the final blueprint.

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OUTCOMES AND EVIDENCE

From building solutions to architecting outcomes

TURNING PRINCIPLES INTO ARCHITECTURE PRACTICE · FIVE HANDS-ON LABS

Multi-model data design
Entity modelling, performance tuning and schema evolution for real workloads

Kafka streaming pipeline
Producers, consumers, transformations and windowing on cloud-native tools

Pipeline orchestration
Automated ingestion with schema mapping and validation in ADF or NiFi

DataOps CI/CD
Git-based versioning, validation tests and IaC deployment

Cloud-native pipeline
Real-time monitoring, query acceleration and cost-efficient storage

Participants moved beyond concepts to design, build and defend working data platforms - modelling for real workloads, building batch and streaming pipelines, automating delivery and tuning for cost and reliability.

THE CAPSTONE · ARCHITECTING COMPLEX DATA SOLUTIONS

MILESTONE 01
Pipeline Implementation
Individual submission: a working pipeline or detailed implementation design, with supporting exercises

MILESTONE 02
End-to-End Architecture Blueprint
A complete architecture design with diagrams, rationale and decision records

WHY A BLUEPRINT, NOT A TEST

Four case-study sessions on complex structured and unstructured datasets from banking, healthcare and media set the context. Teams then designed an architecture demonstrating fault-tolerant design, streaming and batch integration, unstructured data handling and cloud service mappings - and presented the **Architecture Decision Records and trade-offs behind every choice**. A dedicated review-and-refactor session sharpened each design before final submission.

WHAT PARTICIPANTS REPORTED

- Greater confidence in making and defending architectural decisions
- Stronger architectural thinking - a shift from building solutions to architecting outcomes
- A broader perspective on designing enterprise-scale data solutions
- Progression from developing-level proficiency to competent and proficient across key architecture skills

How impact was measured
Proficiency was assessed across 11 competency areas at the start and end of the programme, showing an average gain of +2.0 points. Learner satisfaction ranged from 4.6 to 4.9 out of 5, and the programme recorded a Net Promoter Score of 90.

DEVELOPING · COMPETENT · PROFICIENT

TOOLS AND PLATFORMS

Apache Kafka · Azure Data Factory · Apache NiFi · Microsoft Fabric · Databricks Unity Catalog · Microsoft Purview · Informatica MDM · AWS Glue & Kinesis · GCP Dataflow & BigQuery · Azure DevOps · GitHub Actions · Power BI · draw.io

Platforms and tools explored across Azure, AWS and GCP for architecture patterns, governance, orchestration, DataOps and analytics.

> **The result was more than a training programme - it was architecture capability built from within, with experienced engineers now designing for trade-offs, governance and long-term business outcomes.**

ABOUT STACKROUTE AND RPS CONSULTING

StackRoute and RPS Consulting operate as NIIT's enterprise capability arm, combining advisory, role-based skilling, OEM-certified training, assessment and delivery at scale. The combined entity builds four capability areas for enterprises: **modern engineering; data and AI; architecture and leadership; and execution at scale** - spanning the full talent lifecycle and measured by workforce outcomes rather than training hours.

Outcomes reported at programme completion. Proficiency assessed across 11 competency areas; learner satisfaction rated on a 5-point scale.

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